

SABER: A Semantic-Aligned Brain Network Analysis Framework via Multi-scale Hypergraphs

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Abstract—Effective brain disease diagnosis requires the synergy of brain connectivity patterns and high-level semantic knowledge. Existing methods, however, largely treat semantics from large language models (LLMs) as auxiliary features or supervision, limiting their direct role in decision-making and constraining classification stability and robustness. To overcome this, we propose a semantic-aligned brain network framework that actively integrates LLM-derived semantics into the prediction process. Specifically, ROI-level semantics are first incorporated via global self-attention to enrich node representations and provide whole-brain context. Multi-scale hypergraphs are then constructed to explicitly model functional subnetworks and multi-ROI interactions, addressing the locality limitations of traditional GNNs and capturing high-order dependencies. Finally, a decision-level semantic alignment mechanism selectively injects patient-specific textual embeddings into graph representations, enabling semantics to directly guide predictions without perturbing the underlying network structure. Experiments on public brain network datasets ABIDE and ADHD-200 demonstrate state-of-the-art performance, enhanced stability, and improved interpretability, particularly in small-sample settings.

Index Terms—Hypergraph Neural Networks, Brain Network Analysis, Semantic Alignment.

I. INTRODUCTION

Functional magnetic resonance imaging (fMRI) is a pivotal non-invasive tool for brain network analysis [1]–[3]. By modeling functional connectivity as brain networks, it enables systematic investigation of inter-regional interactions, offering critical diagnostic insights for Autism Spectrum Disorder (ASD) and Attention-Deficit/Hyperactivity Disorder (ADHD). However, complex high-order dependencies, inter-subject variability, and underutilized clinical semantics hinder the extraction of discriminative features.

Since brain networks can be naturally modeled as graph structures, GNNs have emerged as a paradigm for brain modeling. Representative methods like BrainGNN [4] and DHGFormer [5] have advanced the analysis of functional connectivity. To further capture complex interactions, hypergraph-based extensions such as I²HGC [6] have been introduced to model high-order correlations among brain regions. Most graph frameworks overlook the alignment of structural inter-

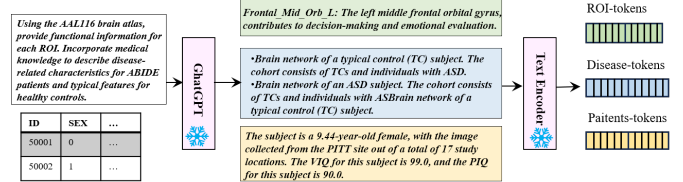


Fig. 1. Overview of the LLM-driven pipeline in our proposed Sabre framework for generating ROI-level, patient-level, and disease-level semantics, serving as structured semantic priors for subsequent brain network analysis.

actions with clinical semantics, failing to leverage semantic intervention for discriminative learning.

In practice, clinicians interpret neuroimaging evidence in conjunction with prior knowledge, such as functional roles of brain regions and disease-related semantics. The emergence of Large Language Models (LLMs) offers a new avenue to encode such high-level knowledge and incorporate it into brain network analysis [7]. Consequently, synergizing high-order brain network modeling with LLM-derived semantics represents an essential research direction.

Two main paradigms currently dominate the integration of fMRI with LLMs. The first involves directly mapping fMRI signals into LLM input tokens for generative tasks, as seen in fMRI-LM [8] and BP-GPT [9]. However, these methods rely heavily on prompt engineering, leading to implicit and often inconsistent decision processes. BP-GPT, for instance, focuses on 1D time series while neglecting high-order interactions among ROIs, treating the LLM merely as a text generator rather than a driver for classification. The second paradigm utilizes LLMs to extract prior knowledge, which is then fused into the imaging pipeline, as in BrainPrompt [10] and LLM-VEM [11]. While BrainPrompt introduces multi-level prompts to enrich ROI, subject, and disease representations, semantic information remains an auxiliary signal, processed only through local GNNs. Consequently, high-level semantics cannot propagate across the whole brain, multi-ROI functional modules are not fully captured, and semantics do not directly participate in decision-making, limiting robustness in small-sample or specific disease tasks.

This limitation raises a fundamental question: can language semantics actively participate in the decision-making process of brain network classification, rather than merely serving

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as feature augmentation or supervision? To address this, we propose Saber, a semantic-aligned brain network modeling framework that elevates LLM-derived semantics to decision-level reasoning components. The overall pipeline for generating ROI-level, patient-level, and disease-level semantics from medical texts using LLMs is illustrated in Figure 5. Specifically, our model integrates ROI semantics via global self-attention at the node level. This enables brain regions to perceive both local connectivity and whole-brain semantic priors, yielding enriched node representations for subsequent high-order modeling. Subsequently, multi-scale hypergraphs are constructed to capture multi-ROI co-activations, overcoming GNN locality and pairwise graph constraints. Meanwhile, a gated fusion mechanism balances cross-scale features to extract robust high-order dependencies. Finally, a graph-level semantic alignment mechanism selectively injects patient-specific textual embeddings into the graph representation, enabling semantic information to directly influence the final prediction without disrupting the original structural information. Our hierarchical design drives multi-scale semantic-structural interactions, improving robustness and discrimination in data-scarce tasks.

The main contributions of this work are summarized as:

- We introduce a decision-level semantic alignment mechanism that enables LLM-derived patient semantics to directly guide brain network classification.
- We propose a multi-scale hypergraph-based node-to-graph modeling strategy to capture high-order and multi-ROI interactions beyond pairwise connectivity.
- Extensive evaluations on ABIDE and ADHD-200 show that Saber consistently outperforms existing methods.

II. METHOD

The overall framework of the Saber is illustrated in Figure 2. In the following, we describe its three stages, progressing from node-level to graph-level representations.

A. Module I: Multi-scale Node-level Brain Network Encoding

Node Feature Initialization. The brain is parcellated into N Regions of Interest (ROIs) utilizing the AAL atlas [12]. We compute the Pearson correlation between the BOLD signals to derive the initial vertex feature matrix $\mathbf{X} \in \mathbb{R}^{N \times N}$.

ROI Semantic Embedding Injection. To incorporate clinical semantic, we generate detailed textual descriptions for each ROI using LLM, focusing on anatomical locations and functional roles. These descriptions are encoded by a frozen pre-trained text encoder to yield semantic embeddings $\mathbf{E}_{\text{ROI}} \in \mathbb{R}^{N \times d_{\text{LLM}}}$. Subsequently, $\mathbf{E}_{\text{ROI}}$ is projected into the node feature space via a learnable linear projection $\mathbf{W}_p$ and fused with the original connectivity-based features $\mathbf{X}$. To facilitate global semantic propagation across regions, we employ the encoder of the Brain Network Transformer (BNT) [13]:

$$\mathbf{X}_0 = \text{BNT}(\mathbf{X} + \mathbf{E}_{\text{ROI}} \mathbf{W}_p), \quad (1)$$

This global self-attention mechanism enables jointly capturing brain connectivity patterns and high-level functional priors, forming the basis for multi-scale hypergraph modeling.

Multi-scale Hypergraph Modeling. Based on the semantic-augmented node features $\mathbf{X}_0$, we construct multi-scale hypergraphs to capture high-order interactions among ROIs. At each scale k , a hypergraph $\mathcal{H}^{(k)} = (\mathcal{V}, \mathcal{E}^{(k)})$ is defined with incidence matrix $\mathbf{H}^{(k)} \in \mathbb{R}^{N \times M_k}$:

$$\mathbf{H}_{i,e}^{(k)} = \begin{cases} 1, & \text{if ROI } i \text{ belongs to hyperedge } e, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where M_k represents the number of hyperedges at scale k . Each hyperedge links a group of ROIs sharing similar semantic-enriched functional characteristics, enabling modeling beyond pairwise connectivity. The ensemble of hypergraphs at multiple scales $\{k_1, \dots, k_K\}$ jointly encapsulates both local and global interaction patterns.

For each scale k , vertex features are propagated through a two-layer Hypergraph Neural Network (HGNN).

$$\mathbf{X}^{(k,l+1)} = \sigma\left(\mathbf{D}_v^{-\frac{1}{2}} \mathbf{H}^{(k)} \mathbf{D}_e^{-1} (\mathbf{H}^{(k)})^\top \mathbf{D}_v^{-\frac{1}{2}} \mathbf{X}^{(k,l)} \mathbf{W}^{(k,l)}\right), \quad (3)$$

where $\mathbf{X}^{(k,0)} = \mathbf{X}_0$, $\mathbf{D}_v$ and $\mathbf{D}_e$ are degree matrices, and $\sigma(\cdot)$ is an activation function. The resulting vertex features at each scale are further refined by a shared Multi-Head Self-Attention (MHSA) module. This module facilitates the transition of information from fine-grained local structures to coarser global patterns while maintaining consistency across scales:

$$\mathbf{Z}^{(k)} = \text{MHSA}\left(\mathbf{X}^{(k,2)}\right). \quad (4)$$

By sharing the attention parameters across scales, the model enforces a unified interaction pattern for integrating multi-scale representations, preventing scale-specific overfitting and enabling coherent aggregation in the subsequent graph-level abstraction module. The resulting multi-scale node representations $\{\mathbf{Z}^{(k)}\}_{k=1}^K$ are then forwarded to Module II.

B. Module II: Graph-level Representation Abstraction

Given the multi-scale node-level representations $\{\mathbf{Z}^{(k)}\}_{k=1}^K$ yielded by Module I, this module synthesizes them into a unified graph-level representation. This process captures subject-specific brain network characteristics while preserving complementary information from varying interaction scales.

In-scale Node-to-Graph Pooling. For each hypergraph scale k , node-level features $\mathbf{Z}^{(k)}$ are aggregated into a scale-specific graph representation via global average pooling:

$$\mathbf{g}^{(k)} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i^{(k)}, \quad (5)$$

where $\mathbf{g}^{(k)}$ summarizes the overall functional organization of the brain network under scale k , providing a compact global descriptor while preserving scale-specific characteristics.

Adaptive Multi-scale Fusion. Recognizing that the contribution of different interaction scales varies across subjects, we employ a learnable gating mechanism to adaptively fuse scale-specific representations. Specifically, the pooled features from all scales are concatenated and passed to a gating network:

$$\alpha = \text{softmax}\left(f_{\text{gate}}\left(\left[\mathbf{g}^{(1)} \parallel \mathbf{g}^{(2)} \parallel \dots \parallel \mathbf{g}^{(K)}\right]\right)\right), \quad (6)$$

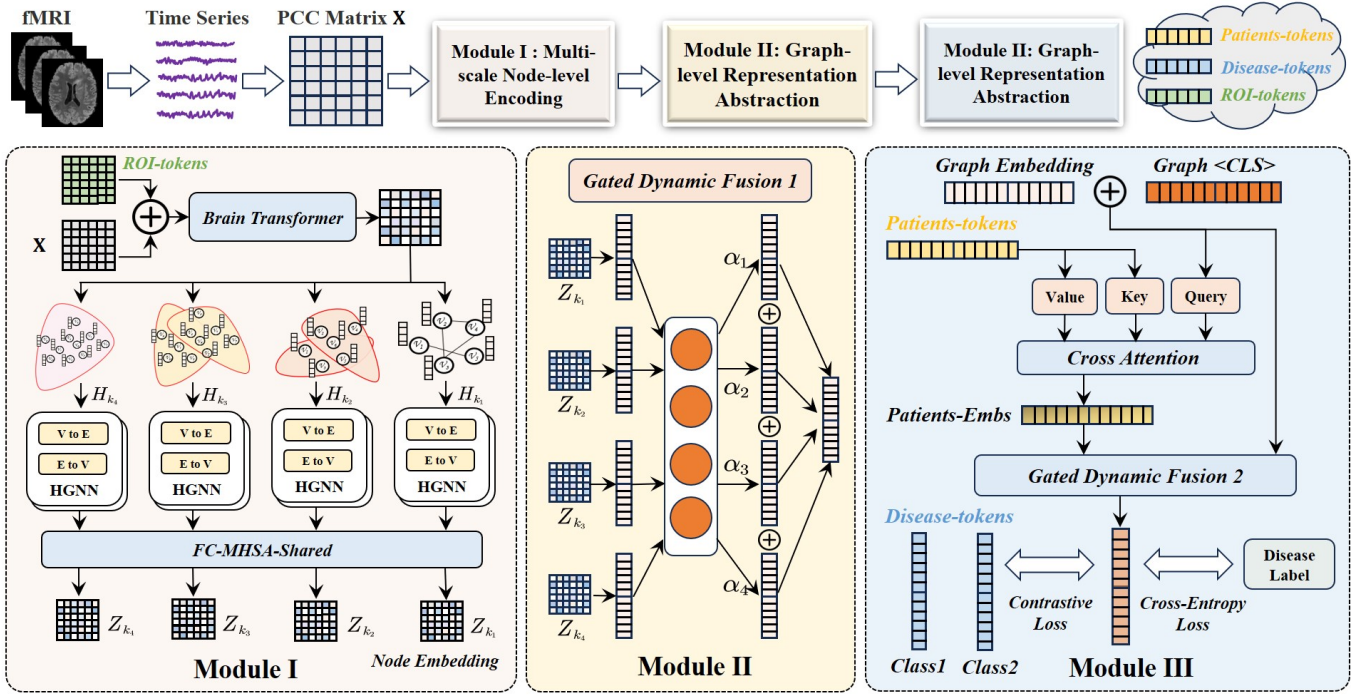


Fig. 2. Overview of the proposed LLM-guided brain network framework Saber that progressively injects semantic priors from ROI-level node representations to graph-level abstraction and decision-level prediction.

where $\alpha \in \mathbb{R}^K$ denotes the subject-specific importance weights for different scales, and $\parallel$ represents the concatenation operation. The final graph-level representation is derived as a weighted combination:

$$\mathbf{g} = \sum_{k=1}^K \alpha_k \mathbf{g}^{(k)}. \quad (7)$$

The resulting representation $\mathbf{g}$ provides a compact yet discriminative summary of the brain network and is forwarded to Module III for decision-level semantic enhancement.

C. Module III: Semantic Alignment and Decision-level Enhancement

Module III advances the integration of LLM-derived semantics from auxiliary feature augmentation to explicit decision-level reasoning. By aligning patient-specific textual information and label semantics with the graph-level representation $\mathbf{g}$, this module allows semantic knowledge to guide predictions while preserving the structural integrity of the brain network. **Patient-specific Semantic Injection.** To enable controlled decision-level reasoning, we introduce a dedicated *Graph-CLS* token $\mathbf{q}_{\text{CLS}} \in \mathbb{R}^d$, which is decoupled from the structural pooling process. Unlike conventional CLS tokens, $\mathbf{q}_{\text{CLS}}$ is fused with the graph-level embedding $\mathbf{g}$ prior to attending to the LLM text, forming a conditioned query:

$$\mathbf{q} = \mathbf{q}_{\text{CLS}} + \mathbf{g}. \quad (8)$$

This design ensures that semantic retrieval is explicitly conditioned on the graph representation. It enables the model

to extract patient-specific semantic cues that are contextually relevant to the brain network, thereby filtering out irrelevant or noisy information, which is a key innovation of this module.

Patient metadata, including demographic and clinical descriptions, are encoded via a pre-trained LLM into a sequence of embeddings $\mathbf{E}_{\text{Patient}} \in \mathbb{R}^{N_{\text{text}} \times d_{\text{LLM}}}$. The combined token $\mathbf{q}$ then attends to these embeddings through cross-attention:

$$\mathbf{h}_{\text{fused}} = \text{CrossAttn}(\mathbf{q}, \mathbf{E}_{\text{Patient}}, \mathbf{E}_{\text{Patient}}), \quad (9)$$

This mechanism allows the model to selectively attend to the most informative patient-specific semantic cues for the prediction task, preventing unfiltered semantic noise from corrupting the graph representation.

Gated Residual Semantic Injection. To regulate semantic contribution, we employ a gated residual fusion mechanism:

$$\mathbf{g}' = \mathbf{g} + \sigma(\mathbf{W}_g \mathbf{h}_{\text{fused}}) \odot \mathbf{h}_{\text{fused}}, \quad (10)$$

where $\sigma(\cdot)$ is sigmoid and $\odot$ is element-wise multiplication. This adaptive gating dynamically modulates the strength of semantic injection per subject, ensuring robustness against noisy or irrelevant textual information.

Decision-level Supervision. The semantically enriched representation $\mathbf{g}'$ is optimized using a dual-supervision strategy. First, standard cross-entropy loss enforces accuracy:

$$\mathcal{L}_{\text{CE}} = -\log \hat{p}_y, \quad \hat{\mathbf{p}} = \text{Softmax}(\text{MLP}(\mathbf{g}')). \quad (11)$$

Second, a contrastive loss aligns $\mathbf{g}'$ with LLM-derived disease embeddings $\mathbf{E}_{\text{Disease}} = \{\mathbf{e}_1, \dots, \mathbf{e}_C\}$, encouraging

proximity to the corresponding class embedding $\mathbf{e}_y$ while pushing away from others:

$$\mathcal{L}_{\text{CL}} = -\log \frac{\exp(\text{sim}(\mathbf{g}', \mathbf{e}_y)/\tau)}{\sum_{c=1}^C \exp(\text{sim}(\mathbf{g}', \mathbf{e}_c)/\tau)}. \quad (12)$$

The total training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CL}}. \quad (13)$$

III. EXPERIMENTS

A. Datasets

1) *ABIDE Dataset*: We evaluated our proposed method on the Autism Brain Imaging Data Exchange (ABIDE I) dataset [20]. ABIDE I consists of rs-fMRI data collected from 16 international sites. To ensure data quality, we selected 871 quality-checked subjects, comprising 403 patients with Autism Spectrum Disorder (ASD) and 468 Normal Controls (NC).

2) *ADHD-200 Dataset*: We further validated our model using the ADHD-200 dataset, a large-scale public database aggregated from eight imaging centers. This dataset includes 362 patients diagnosed with Attention Deficit Hyperactivity Disorder (ADHD) and 585 NCs. Both datasets were pre-processed using the DPARSF pipeline [21].

To ensure robust training and evaluation, we employed a stratified sampling strategy to partition the dataset into training, validation, and test sets with an 8:1:1 ratio. This stratification preserves the class distribution across splits, ensuring a balanced representation of patients and normal controls.

B. Evaluation Setups

1) *Implementation Details.*: Model training was performed on a single NVIDIA RTX 3090 GPU with 24GB of memory. The batch size was set to 16, and the model was trained for 100 epochs using the AdamW optimizer. We adopted an initial learning rate of 1×10^{-4} and applied weight decay of 1×10^{-4} to prevent overfitting and enhance generalization. A cosine annealing learning rate scheduler was used during training to improve convergence. To further validate the model's reliability and generalizability, we also conducted ten-fold cross-validation.

For text generation and encoding, we use ChatGPT-5 and Llama-encoder-1.0B model from the LLM2Vec framework as a frozen encoder. ROI, patients, and disease text are encoded in a d_{llm} dimensional embedding, which is then linearly projected onto a 116-dimensional vector matching the node features. We evaluated hypergraphs across multiple scales ($k \in \{1, 5, 10, 15\}$) to capture multi-resolution structural representations of brain dynamics. These localized and global features are fused within a unified framework. Based on empirical performance, we set the number of attention heads to 8.

2) *Baseline Methods*: We evaluated the proposed model against several state-of-the-art baselines, including BrainNetCNN [14], HGNN [16], BrainNetTransformer [13], BrainGNN [4], Com-BrainTF [15], MADEforASD [19], BrainGSL [18] and BrainPrompt [10]. These methods represent a diverse range of brain network modeling strategies.

3) *Evaluation Metrics*: Model performance was assessed using Accuracy, AUC, Recall, Specificity, and F1-score to provide a comprehensive multi-perspective evaluation.

C. Classification Performance Results and Analysis

As illustrated in Table I, the proposed Saber model achieves state-of-the-art performance on both ABIDE (attaining 71.58 accuracy) and ADHD-200 datasets. These results validate our core hypothesis: elevating clinical semantics from auxiliary features to decision-level components significantly bolsters the robustness of brain network analysis.

1) *Structural Superiority*: Compared to Transformer-based architectures such as BrainNetTransformer and Com-BrainTF, Saber's advantage stems from its Multi-scale Hypergraph Modeling (Module I). While standard Transformers struggle to capture the non-Euclidean topology of brain networks, our approach extracts high-order interactions that align more closely with the brain's intrinsic functional organization. Notably, even the Saber_{lite} (the version without LLM enhancement) consistently outperforms most baselines, including self-supervised frameworks like BrainNPT, BrainGSL.

2) *Synergy of Semantic-Decision Alignment*: Saber demonstrates a substantial leap in classification accuracy and stability over BrainPrompt. While BrainPrompt primarily treats semantics as local GNN inputs, Saber's Graph-CLS Token acts as a decision-level query, selectively extracting critical evidence from patient-specific texts via cross-attention. This design directly addresses the fundamental limitation where semantics fail to participate in the final decision-making process. Through Gated Residual Injection, clinical text evolves from a passively attached label into an active constraint that shapes the classification boundary. The consistent performance across both datasets highlights Saber's exceptional adaptability; by synergizing high-order imaging features with textual priors, the model effectively emulates the "imaging + clinical records" dual-verification logic employed by medical experts.

D. Ablation Study

Table II shows ablation results on ABIDE and ADHD-200.

1) *Multi-scale structure & shared attention*: Removing the Transformer or using a single scale ($k = 10$) degrades performance, indicating that fixed-scale modeling fails to capture high-order brain interactions. We therefore adopt a multi-scale setting with $k \in \{1, 5, 10, 15\}$ to hierarchically model structures from local ROIs to large-scale inter-regional coordination. In contrast, disabling shared attention (MHSA-noshared) disrupts cross-scale alignment and reduces performance consistency.

E. Impact of Key Hyperparameters on Model Performance

1) *Dynamic Fusion & Graph-CLS*: Excluding dynamic fusion weakens performance, indicating adaptive multi-scale integration is crucial. Removing Graph-CLS or both CLS and dynamic fusion further lowers results, showing decision-level queries aid semantic selection without harming structural representation.

TABLE I
COMPARISON ON ABIDE AND ADHD DATASETS. BEST RESULTS ARE **BOLDED**. ALL VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION (%).
SABER_{LITE} DENOTES THE VARIANT WITHOUT LLM-BASED SEMANTIC MODELING, WHILE SABER REPRESENTS THE FULL MODEL.

Method	ABIDE Dataset					ADHD Dataset	
	Accuracy	AUC	Recall	Specificity	F1 Score	Accuracy	AUC
BrainNetCNN [14]	65.81 $\pm$ 1.91	70.22 $\pm$ 2.66	69.03 $\pm$ 3.96	65.97 $\pm$ 2.51	67.33 $\pm$ 1.25	61.42 $\pm$ 1.87	62.03 $\pm$ 2.14
BrainGNN [4]	64.43 $\pm$ 1.66	68.14 $\pm$ 1.69	65.31 $\pm$ 5.44	63.81 $\pm$ 3.79	64.41 $\pm$ 3.69	60.95 $\pm$ 2.02	61.88 $\pm$ 2.36
BNT [13]	66.13 $\pm$ 2.36	71.34 $\pm$ 3.95	71.87 $\pm$ 6.96	67.41 $\pm$ 3.71	69.33 $\pm$ 3.31	62.87 $\pm$ 1.74	63.45 $\pm$ 2.01
Com-BrainTF [15]	64.41 $\pm$ 2.89	70.77 $\pm$ 2.85	69.03 $\pm$ 6.12	66.35 $\pm$ 3.22	67.44 $\pm$ 3.16	62.11 $\pm$ 2.28	62.74 $\pm$ 2.47
HGNN [16]	64.52 $\pm$ 2.10	70.41 $\pm$ 2.37	66.72 $\pm$ 5.81	65.15 $\pm$ 3.90	65.67 $\pm$ 2.59	61.76 $\pm$ 1.93	62.20 $\pm$ 2.10
BrainNPT [17]	64.32 $\pm$ 3.30	69.47 $\pm$ 4.33	66.39 $\pm$ 4.11	65.38 $\pm$ 4.97	65.57 $\pm$ 0.81	61.58 $\pm$ 2.41	62.09 $\pm$ 2.68
BrainGSL [18]	66.31 $\pm$ 4.38	67.35 $\pm$ 5.55	66.82 $\pm$ 5.58	67.57 $\pm$ 5.57	66.93 $\pm$ 3.87	63.24 $\pm$ 2.35	63.88 $\pm$ 2.79
MADEforASD [19]	65.21 $\pm$ 2.28	69.49 $\pm$ 1.28	69.69 $\pm$ 4.77	65.09 $\pm$ 3.80	67.13 $\pm$ 2.22	62.63 $\pm$ 1.96	63.14 $\pm$ 2.22
BrainPrompt [10]	66.35 $\pm$ 7.14	66.45 $\pm$ 6.55	71.08 $\pm$ 13.42	68.38 $\pm$ 8.54	68.72 $\pm$ 8.11	64.92 $\pm$ 1.88	65.37 $\pm$ 2.04
SABER _{lite}	69.82 $\pm$ 1.34	72.23 $\pm$ 1.15	71.97 $\pm$ 4.83	68.88 $\pm$ 4.14	70.02 $\pm$ 3.06	65.12 $\pm$ 2.42	65.68 $\pm$ 2.77
SABER	71.58 $\pm$ 4.08	70.40 $\pm$ 3.02	81.87 $\pm$ 3.02	69.68 $\pm$ 6.25	75.06 $\pm$ 5.82	66.21 $\pm$ 2.31	66.87 $\pm$ 2.59

TABLE II
ABLATION STUDY ON DIFFERENT ARCHITECTURAL COMPONENTS.

Configuration	ABIDE		ADHD-200	
	Acc	AUC	Acc	AUC
w/o TransEnc	61.84	66.75	60.92	63.41
single-scale (k=10)	61.36	64.84	60.15	62.87
double-scale (k=5,10)	64.22	69.25	63.01	66.38
MHSA-noshared	63.92	68.97	62.48	65.92
w/o Dynamic Fusion	64.02	68.82	62.77	65.41
w/o Dynamic Fusion & CLS	64.72	69.74	63.35	66.02
w/o CLS	63.43	69.52	62.11	65.76
w/o ROI Semantic Injection	70.28	71.43	65.17	65.59
w/o Patient Semantic Injection	70.11	69.89	64.98	65.09
w/o Disease Contrastive	68.47	68.94	64.12	64.79
SABER _{lite}	69.82	72.23	65.12	65.68
SABER	71.58	70.40	66.21	66.97

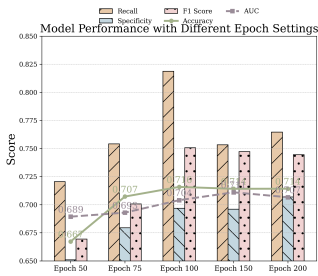


Fig. 3. Perf. vs. Epochs.

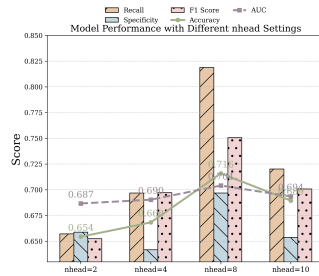


Fig. 4. Effect of Heads.

2) *Hierarchical semantic alignment*: Removing ROI- or patient-level semantic injection reduces performance, and omitting disease-level semantic contrast significantly harms discriminative power, confirming the complementary role of multi-level semantic priors.

Full SABER consistently outperforms SABER_{lite}, demon-

TABLE III
PERFORMANCE COMPARISON UNDER DIFFERENT BATCH SIZES (%)

Batch Size	Accuracy	AUC	Recall	Specificity	F1
4	62.38	67.10	66.35	62.73	64.49
8	65.89	67.25	73.23	65.50	66.57
16	71.58	70.04	81.87	69.68	75.06
32	64.85	66.04	70.19	64.60	67.28
64	63.37	66.33	72.12	62.50	66.96

strating that decision-level semantic enhancement stabilizes performance and improves generalization.

We also evaluated different training durations (50–200 epochs). As shown in Figure 3, performance peaks at 100 epochs, with longer training causing slight declines, likely due to overfitting. This identifies 100 epochs as the optimal setting.

The number of attention heads was fixed across Transformer layers for consistency. Ablation studies (Figure 4) show performance improves from 2 to 8 heads but slightly drops at 10, indicating that a moderate head count balances feature diversity and stability.

We evaluated the effect of different batch sizes on model performance (Table III). Smaller batches improved sensitivity, while larger batches enhanced training stability, each with trade-offs in efficiency or generalization. Balancing performance and training time, we chose a batch size of 16 for optimal efficiency and accuracy.

F. Interpretability Analysis of Discriminative ROIs

To improve model interpretability and explore its neuroscientific relevance, we analyzed the most discriminative functional connections and ROIs for ASD classification. Using individualized intra-hypergraph representations, we constructed subject-specific FC matrices. Key connections were identified via a combination of two-sample t-tests and Lasso regression, ensuring both statistical significance and sparsity.

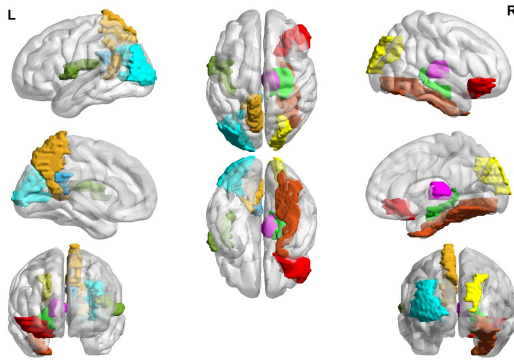


Fig. 5. The visualization on discriminative ROIs for ASD Diagnosis.

The analysis highlighted that the most discriminative connections predominantly involve the fusiform gyrus, hippocampus, inferior frontal gyrus (orbital part), middle occipital gyrus, and posterior cingulate cortex. These regions, frequently implicated in significant connections, may serve as potential neurobiological biomarkers for ASD, consistent with prior findings in the literature [22] (see Fig. 5).

IV. CONCLUSION

In this paper, we proposed Saber, a semantic-aligned brain network modeling framework that integrates LLM-derived textual knowledge with multi-scale fMRI-based hypergraph representations. By progressively injecting semantic priors from ROI-level node features to graph-level abstraction and decision-level prediction, Saber enables patient-specific semantics to actively guide classification. Multi-scale hypergraph modeling captures high-order, cross-ROI interactions beyond pairwise connectivity, while adaptive fusion and gated semantic injection ensure robust integration of structural and semantic information. Extensive experiments on ABIDE and ADHD-200 demonstrate that Saber consistently outperforms existing methods, offering improved accuracy, robustness, and interpretability. Furthermore, analysis of discriminative functional connections and ROIs reveals neurobiologically meaningful patterns, highlighting the potential of combining high-order brain network modeling with LLM-driven semantic knowledge for advancing neurodevelopmental disorder analysis.

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